Interview notes

Speak to evidence, assumptions and limits | Executed portfolio project

Business framing
I used 307,511 Home Credit applications as a public consumer-credit proxy. It demonstrates MSME-relevant methods, but does not establish performance on Indian merchants.

Why SQL aggregation?
I collapsed each child history to applicant level before joining, preventing one-to-many row multiplication. The final table has 168 candidate features; IDs and TARGET are excluded from that count.

Why random holdout?
There is no clean application date and IDs are not proven chronological. I used disjoint stratified random train, policy, test and holdout partitions; I do not claim out-of-time validation.

Why WoE and logistic?
WoE turns bins into smoothed log good/bad ratios. I fit bins, IV screening and correlation/VIF checks on training only, retaining 20 features for an additive, auditable logistic scorecard.

What do AUC and Gini mean?
AUC is the chance a random bad applicant ranks riskier than a random good. Gini = 2 x AUC - 1; holdout scorecard Gini is 0.5039, versus 0.5626 for LightGBM.

What is KS?
KS is the maximum difference between cumulative bad and good distributions across risk thresholds. Holdout scorecard KS is 37.64%; it measures separation, not probability calibration.

How do points work?
The configured base is 600 points at 50:1 good/bad odds. Each 20 points doubles those odds; I verified exact unrounded point additivity.

Calibration and PSI
Calibration compares predicted and observed risk; I report Brier scores and reliability curves without fitting a calibrator. Holdout score PSI is 0.00016; <0.1 stable, 0.1-0.25 moderate, >0.25 shifted are heuristic alerts.

Why boosting and SHAP?
LightGBM captures nonlinear interactions. Candidate settings use training-only cross-validation; tree-path-dependent SHAP explains contributions to log odds, with a numerical additivity check. Explanations are associative.

How was the cut-off chosen?
I optimized assumed net benefit on the policy partition with a minimum approval constraint, then froze score 545. Holdout estimated bad-rate reduction is 38.59%; good rejection share is 18.08%.

How were rupee estimates made?
With assumed 100,000 annual applications, Rs 500,000 loans, 60.00% LGD and 8.00% good margin, gross avoided loss is Rs 124.34 crore; net benefit is Rs 57.86 crore. These are illustrative, not realized.

Limitations and SIMULATED extension
No temporal validation, verified MSME population, causal policy effect or fairness certification is claimed. SIMULATED sales recall is 100.00%, with 16 SIMULATED days median lead time; built-in decline drives that artificial result.